USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1. Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2. For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3. Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1.b)
4. Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: TEST.PD.EPR.SNK3.13
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:
The current CTS covers SPR-mode PDO transition behavior for Sink UUTs in **TEST.PD.PS.SNK.1 PDO Transitions As Sink**, but provides **no equivalent test for EPR-mode PDO transitions**. This ECR proposes a new test TEST.PD.EPR.SNK3.13 EPR PDO Transitions As Sink that mirrors TEST.PD.PS.SNK.1's structure but operates against the EPR contract and EPR PDO/APDO set

Benefits as a result of the proposed changes:
Closes a normative-coverage gap. Verifying voltage-transition behavior is essential for interop with the new generation of 28V/36V/48V Fixed PDOs and 15V-48V EPR AVS.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
Tester to implement the changes per ECR update.

An analysis of the Vendor Info File (VIF) implications:
None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

TEST.PD.EPR.SNK3.13 EPR PDO Transitions As Sink

To Text:

TEST.PD.EPR.SNK3.13 EPR PDO Transitions As Sink

Description: The Tester verifies that an EPR-capable Sink UUT, after entering EPR Mode, correctly responds to new EPR_Source_Capabilities Messages, requests valid EPR PDO/APDOs per its VIF, and that VBUS transitions stay within the EPR voltage envelopes.

Test Specific Tester Behavior:

Test Conditions:

This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓
Rev3UnchkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

   a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

   b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. The Tester sends a *Get_Sink_Cap_Extended* Message to the UUT, the check fails if the UUT does not send a *Sink_Capabilities_Extended* Message. [TEST.PD.EPR.SNK3.13#1]

3. The Tester waits up to 5s for the UUT to send an *EPR_Mode* (Enter, Action = 0x01) Message. If the VIF field EPR_Supported_As_Snk is Yes, the test stops here with a warning if no *EPR_Mode* Enter Message is received. [TEST.PD.EPR.SNK3.13#2]

4. If VIF field Captive_Cable = Yes, the Tester sends SOP' *Discover Identity* Command and checks the below fields in the response message, [TEST.PD.EPR.SNK3.13#3]
   a. V_{BUS} Current Handling Capability set to 10b(5A)

b. Maximum V_{BUS} Voltage set to 11b(50V)
c. EPR Mode Capable field set to 1b (EPR Mode Capable)

5. The Tester sends the *EPR_Mode* Message with Action field set to 0x03(Enter_Succeeded) and the Data field set to 0x0 within *tEnterEPR* from the *EPR_Mode* (Enter) Message.

6. The Tester sends the *EPR_Source_Capabilities* Message after 50ms from the last bit of the *GoodCRC* Message acknowledging the *EPR_Mode* (Enter_Succeeded) Message and establishes EPR contract using common procedure COMMON.PROC.PD3.3 from Step2. The Source_Capabilities Extended Message (sent on Get_Source_Cap_Extended) reports SPR Source PDP Rating = 100W and EPR Source PDP Rating = 240W. [TEST.PD.EPR.SNK3.13#4]

7. The Tester checks that the requested PDO/APDO is within the range of VIF Sink Capabilities Extended PDP for EPR mode: [TEST.PD.EPR.SNK3.13#5]

a. For an EPR Fixed *EPR_Request* Message, the check fails if B19…10(Operating Current)*Voltage of the referenced PDO is greater than VIF field Snk_EPR_Maximum_PDP.

b. For an EPR_AVS *EPR_Request* Message, the check fails if:
   i. Snk_AVS_Supported = No and EPR_AVS *EPR_Request* is sent
   ii. Output Voltage is outside [15V, max offered EPR Fixed voltage], or
   iii. B20…9 (Output Voltage)*B6…0(Operating Current) is greater than VIF field Snk_EPR_Maximum_PDP.

8. The Tester intentionally sends a *Wait* Message. The test fails if another *EPR_Request* Message is received before *tSinkRequest* min. This is measured when the EOP of the *GoodCRC* acknowledging the *Wait* Message. [TEST.PD.EPR.SNK3.13#6]

9. If the UUT does not resend the *EPR_Request* Message within *tCtsInfinite*, the Tester resends the *EPR_Source_Capabilities* Message per step 6. The test fails if the UUT does not respond with an *EPR_Request* Message to the resent *EPR_Source_Capabilities*. [TEST.PD.EPR.SNK3.13#7]

10. The Tester sends an *Accept* Message to the *EPR_Request* Message received.

11. If the *EPR_Request* asks for an increasing-voltage transition:

a. If the transition is to an EPR Fixed Supply PDO, or to the EPR AVS APDO with a voltage step greater than *vSmallStep*:

   i. Within *tSnkStdby* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that current drawn by the UUT does not exceed *iSnkStdby* max. [TEST.PD.EPR.SNK3.13#8]

ii. Wait a nominal *tSrcTransition* from the *GoodCRC* and then change V_{BUS} voltage to the new requested value. For EPR Fixed and EPR AVS large-step transitions, the voltage change shall complete before *tSrcTransReq* expires.

iii. As soon as the volage on V_{BUS} has stopped changing, check that current drawn by the UUT does not exceed *iSnkStdby* max until *PS_RDY* is sent. [TEST.PD.EPR.SNK3.13#9]

b. If the transition is to EPR AVS with a voltage step less than or equal to vSmallStep:

i. The Tester shall immediately begin changing V_{BUS} voltage after the *GoodCRC* Message acknowledging the *Accept* Message. The tester shall complete the voltage transition and send *PS_RDY* before *tAvsSrcTransSmall* expires.

ii. If the newly negotiated current is less than the previously contracted current, after *tSnkNewPower* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that the current drawn by the UUT does not exceed the newly contracted current (I_{NEW}). [TEST.PD.EPR.SNK3.13#10]

iii. If the newly negotiated current is unchanged or increased, check that the current drawn by the UUT does not exceed the previously contracted current during the transition after the EOP of *GoodCRC* acknowledging the *Accept* Message until the *PS_RDY* is sent by the Tester. [TEST.PD.EPR.SNK3.13#11]

11. If the *EPR_Request* asks for an unchanged or decreasing voltage, or no new PDO is requested:
    a. If the newly negotiated current is unchanged or increased: after *tSrcTransReq* EPR Mode max time period (before *PS_RDY* is sent) from the *GoodCRC* acknowledging the *Accept* Message, check that the current drawn from V_{BUS} does not exceed the previously contracted current. [TEST.PD.EPR.SNK3.13#12]
    b. If the newly negotiated current is decreased: after *tSnkNewPower* from the EOP of GoodCRC acknowledging the Accept Message, check that the current drawn by the UUT does not exceed I_{NEW}. [TEST.PD.EPR.SNK3.13#13]

12. The Tester sends a *PS_RDY* Message to the UUT.

13. The Tester checks that the current drawn by the UUT does not exceed the level specified in the contracted PDO over the next 5 seconds. [TEST.PD.EPR.SNK3.13#14]

14. The Tester offers an *EPR_Source_Capabilities* Message with EPR PDOs removed from the initially offered *EPR_Source_Capabilities* PDOs in Step 6 with a reduced Maximum Voltage of 20V and Port Maximum PDP of 100W. The check fails if the UUT does not send an *EPR_Request* Message for an SPR-only PDO. If *EPR_Request*

voltage is not same as the current contract, repeat steps 7-10. If the requested voltage is same as the current contract, test ends here. [TEST.PD.EPR.SNK3.13#15]

15. The Tester re-offers an *EPR_Source_Capabilities Message* with the Maximum Voltage of 48V as initially offered in Step 6. If *EPR_Request* voltage is not same as the current contract, repeat steps 7-10. If the requested voltage is same as the current contract, test ends here. [TEST.PD.EPR.SNK3.13#16]